

E-Commerce Customer Lifecycle & Value (LTV) Prediction

Using RFM Analysis and Machine Learning

Submitted by:

B.R. Bavadharani

B.Sc. Economics (Hons.) with Specialization in Data Science

Vellore Institute of Technology (VIT), Chennai

Dantuluri Bhanu Teja Varma

B.Tech in Computer Science and Engineering (CSE)

MVGR College of Engineering

Project Guide / Mentor Name:

Mr. Arpan Roy

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1. ABSTRACT

The rapid growth of e-commerce platforms has resulted in the generation of massive volumes of customer transaction data. These data contain valuable insights into customer purchasing behaviour, spending habits, and shopping patterns. Effectively analysing customer behaviour enables organisations to improve marketing strategies, strengthen customer relationships, and maximise long-term profitability.

This project, titled "Customer Segmentation and Customer Lifetime Value (LTV) Prediction using RFM Analysis and Machine Learning" focuses on understanding customer purchasing behaviour through data analytics and predictive modelling using the UCI Online Retail dataset.

The project begins with comprehensive data preprocessing, including the removal of missing customer identifiers, cancelled invoices, duplicate records, and invalid transactions. A Revenue variable was then created by multiplying product quantity with unit price. Following data cleaning, Recency, Frequency, and Monetary (RFM) metrics were generated to convert transaction-level data into customer-level behavioural information.

Exploratory Data Analysis (EDA) was conducted using descriptive statistics, histograms, and boxplots to understand the distribution of customer behaviour and identify outliers. Since the Frequency and Monetary variables exhibited significant right-skewness, logarithmic transformation was applied using NumPy's `log1p()` function to reduce skewness. Subsequently, StandardScaler was used to standardize all RFM variables before clustering.

Customer segmentation was performed using the K-Means clustering algorithm. The optimal number of clusters was determined using the Elbow Method, resulting in four meaningful customer groups: VIP Customers, Loyal Customers, Regular Customers, and At-Risk Customers.

The final stage of the project involved Customer Lifetime Value (LTV) prediction using three regression algorithms: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Among the evaluated models, Linear Regression achieved the best performance for the selected feature set.

The project demonstrates how machine learning techniques can transform raw transactional data into actionable business insights, enabling organisations to improve customer retention, personalise marketing campaigns, and support strategic decision-making.

2. INTRODUCTION

The rapid growth of e-commerce has generated vast amounts of customer transaction data, providing businesses with valuable insights into purchasing behaviour, spending patterns, and customer preferences. Analysing these data enables organisations to better understand customer behaviour, improve marketing strategies, and make informed business decisions.

Traditional marketing approaches often treat all customers alike, resulting in inefficient resource utilisation and reduced customer engagement. To overcome this challenge, organisations increasingly employ customer analytics and machine learning techniques to identify groups of customers with similar purchasing behaviour and develop personalised marketing strategies. Consequently, customer segmentation has become an important component of Customer Relationship Management (CRM), helping businesses improve customer retention and maximise long-term profitability.

2.1. Customer Segmentation

Customer segmentation is the process of grouping customers with similar purchasing characteristics to enable targeted marketing and improved customer relationship management. In this project, segmentation is performed using **RFM Analysis** and the **K-Means clustering algorithm**.

The RFM model evaluates customer behaviour using three key indicators:

- **Recency (R):** Number of days since the customer's most recent purchase.
- **Frequency (F):** Number of unique purchase transactions made by the customer.
- **Monetary (M):** Total revenue generated by the customer.

These metrics provide a comprehensive view of customer purchasing behaviour and form the basis for customer segmentation.

2.2. Customer Lifetime Value (LTV)

Customer Lifetime Value (LTV) represents the total value a customer contributes to a business throughout their relationship with the organisation. Estimating LTV helps businesses identify high-value customers, improve customer retention, optimise marketing expenditure, and develop effective loyalty programmes.

In this study, the **Monetary** component of RFM analysis is used as a proxy for Customer Lifetime Value, and machine learning regression models are employed to predict customer value.

2.3. Machine Learning in Customer Analytics

Machine learning techniques enable businesses to identify hidden patterns in customer data and generate accurate predictions. This project combines both **unsupervised** and **supervised** learning approaches.

- **K-Means Clustering** is used to classify customers into meaningful segments based on purchasing behaviour.
- **Linear Regression, Decision Tree Regressor, and Random Forest Regressor** are developed to predict Customer Lifetime Value.

The integration of customer segmentation and predictive modelling provides valuable insights for customer relationship management and data-driven decision-making.

Training Topics Covered (First Two Weeks of Internship)

During the initial phase of the internship, training sessions were conducted to strengthen fundamental knowledge in data science and machine learning. The topics covered included:

- Introduction to Data Science
- Basic Statistics for Data Science
- Data Visualization
- Introduction to GitHub and Cloud Computing
- Data Science Application Development using Streamlit
- Machine Learning – I
- Machine Learning – II
- Large Language Model (LLM) Fundamentals
- Communication Skills
- Deep Learning – I

Problem Statement:

Retail organisations generate large volumes of customer transaction data, but extracting meaningful insights from these data remains a significant challenge. Generic marketing strategies often fail to address variations in customer purchasing behaviour, reducing their overall effectiveness. Furthermore, identifying customers with high long-term value is essential for improving customer retention and resource allocation.

This project addresses these challenges by applying **RFM analysis**, **K-Means clustering**, and **machine learning regression models** to segment customers and predict Customer Lifetime Value using real-world e-commerce transaction data.

Scope of the Study:

This study focuses on analysing customer transaction data from the **UCI Online Retail dataset** to understand purchasing behaviour, perform customer segmentation, and predict Customer Lifetime Value using Python-based machine learning techniques. The proposed framework demonstrates how customer analytics can support personalised marketing, customer retention, and data-driven decision-making in the e-commerce sector.

3. OBJECTIVES

The present study was undertaken with the following objectives:

1. **To prepare and refine** the Online Retail dataset by eliminating incomplete, duplicate, and invalid transaction records, thereby ensuring the quality and reliability of data for subsequent analysis.
2. **To examine customer purchasing patterns** through Exploratory Data Analysis (EDA) and RFM (Recency, Frequency, and Monetary) analysis to gain meaningful insights into customer behaviour and spending trends.
3. **To preprocess customer behavioural data** by applying appropriate transformation and scaling techniques, enabling effective implementation of clustering algorithms.
4. **To classify customers into meaningful segments** using the K-Means clustering algorithm and evaluate the optimal number of clusters through the Elbow Method for improved customer profiling.
5. **To estimate Customer Lifetime Value (LTV)** by developing and comparing multiple regression models, namely Linear Regression, Decision Tree Regressor, and Random Forest Regressor, and selecting the most suitable model based on predictive performance measures.
6. **To demonstrate the practical application of customer analytics** in supporting personalised marketing, customer retention strategies, and informed business decision-making within the e-commerce domain.

The project is based entirely on secondary data obtained from the UCI Online Retail dataset. No primary survey or hypothesis-testing procedures were undertaken as part of this study.

4. Methodology

4.1. Research Methodology

This project follows a structured data analytics and machine learning methodology to analyse customer purchasing behaviour and predict Customer Lifetime Value (LTV). The methodology begins with collecting a real-world e-commerce transaction dataset, followed by data cleaning and preprocessing to improve data quality. Customer behavioural features were then generated using RFM (Recency, Frequency, and Monetary) analysis. Exploratory Data Analysis (EDA) was carried out to understand the distribution of customer behaviour and identify outliers.

Since the RFM variables exhibited skewed distributions, logarithmic transformation and feature standardisation were applied to prepare the data for clustering. Customer segmentation was performed using the K-Means clustering algorithm, with the optimal number of clusters determined using the Elbow Method. Finally, three regression models—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—were developed to predict Customer Lifetime Value. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The overall methodology adopted in this project is illustrated in the figure below.

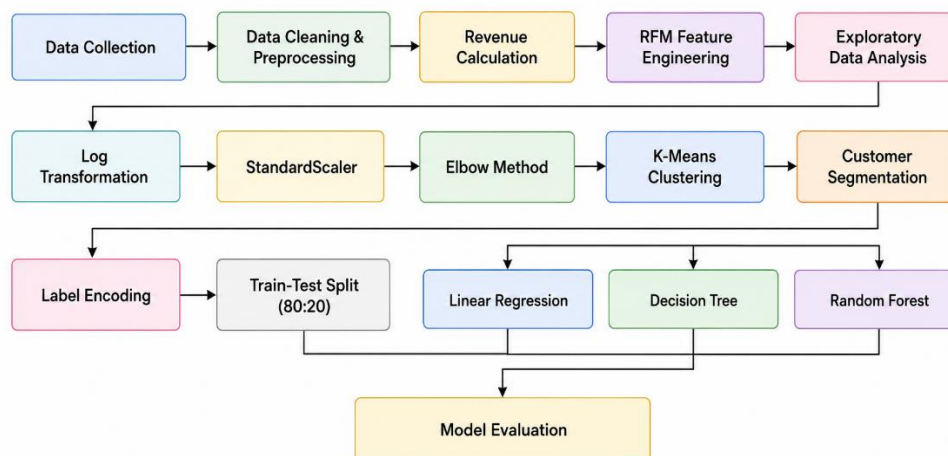


Figure 1- Methodology

4.2. Data Source and Description

The study is based entirely on secondary data obtained from the UCI Machine Learning Repository. The dataset used is the Online Retail Dataset, which contains transactional records of a UK-based online retail company specialising in gift

products. The dataset includes transactions recorded between 1 December 2010 and 9 December 2011, covering customers from multiple countries.

The original dataset consists of 541,909 transaction records and 8 variables. Each row represents a single product purchased within a customer invoice. Since the project focuses on customer behaviour analysis rather than geographic analysis, all available records were considered during preprocessing.

The variables contained in the dataset are presented in Table 4.1.

Table 4.1 Dataset Variables

Variable	Description
InvoiceNo	Unique invoice number assigned to each transaction
StockCode	Product identification code
Description	Name of the purchased product
Quantity	Number of product units purchased
InvoiceDate	Date and time of the transaction
UnitPrice	Price per unit (GBP)
CustomerID	Unique customer identifier
Country	Customer's country

The Online Retail dataset was selected because it is one of the most widely used benchmark datasets for customer analytics, customer segmentation, and machine learning research. The presence of customer identifiers, transaction dates, purchase quantities, and product prices makes it particularly suitable for RFM analysis and Customer Lifetime Value prediction.

4.3. Software and Tools Used

The project was implemented entirely in the Python programming environment using Jupyter Notebook. Python was selected because of its extensive ecosystem of libraries for data manipulation, statistical analysis, machine learning, and visualisation. It was then uploaded to GitHub repository.

Table 4.2 Software and Libraries Used

Software / Library	Purpose
Python	Programming language
Jupyter Notebook	Development environment
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
Matplotlib	Data visualisation
Seaborn	Statistical visualisation
Scikit-learn	Machine learning algorithms
Microsoft Excel	Preliminary data inspection

4.4. Data Cleaning and Preprocessing

Raw transactional datasets frequently contain incomplete, duplicate, and invalid records that can adversely affect analytical results. Therefore, a systematic data preprocessing procedure was carried out before performing customer segmentation and predictive modelling.

The following preprocessing steps were implemented.

Step 1: Importing the Dataset

The Online Retail dataset was imported into Python using the Pandas library. The dataset dimensions and variable types were examined to understand its overall structure before further processing.

Step 2: Removal of Missing Customer IDs

Some transaction records contained missing values in the CustomerID field. Since customer-level analysis requires a unique identifier, these records were removed using the `dropna()` function.

This step ensured that every transaction could be associated with a valid customer during RFM analysis.

Step 3: Removal of Cancelled Transactions

Cancelled transactions were identified through invoice numbers beginning with the letter "C". These invoices represent returned or cancelled purchases and therefore do not represent actual customer spending.

All cancelled invoices were removed from the dataset to ensure that only completed purchase transactions were analysed.

Step 4: Removal of Invalid Quantity Values

Transactions with quantities less than or equal to zero were excluded. Negative quantities generally represent product returns, while zero quantities do not correspond to valid purchases.

Removing these records ensured that customer purchasing behaviour reflected only genuine sales transactions.

Step 5: Removal of Invalid Unit Prices

Records containing zero or negative unit prices were removed because they do not represent valid commercial transactions and would lead to incorrect revenue calculations.

Step 6: Removal of Duplicate Records

Duplicate transaction records were identified using the duplicated () function and removed from the dataset. Eliminating duplicate observations prevented overestimation of customer purchase frequency and total expenditure.

Step 7: Date Conversion

The InvoiceDate variable was converted to Python's datetime format using the pd.to_datetime () function. This conversion enabled accurate time-based calculations required for computing customer Recency.

Step 8: Customer ID Conversion

The CustomerID variable was converted from floating-point format to integer format to simplify customer-level aggregation and grouping operations.

Step 9: Revenue Calculation

A new variable named Revenue was created using the following expression:

$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$

This variable represents the monetary value generated by each transaction and forms the basis for the Monetary component of RFM analysis.

Step 10: Data Validation

After preprocessing, the cleaned dataset was validated by checking for:

- Missing Customer IDs
- Duplicate records
- Invalid quantity values
- Invalid unit prices

The validation confirmed that the dataset was free from major inconsistencies and suitable for customer analytics and machine learning applications.

4.5. Feature Engineering

Feature engineering was carried out to transform transaction-level data into customer-level behavioural data suitable for segmentation and predictive modelling.

Revenue: Revenue was calculated for every transaction as:

$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$

This variable represents the monetary contribution of each transaction.

RFM Analysis

Customer behaviour was summarised using three key indicators:

- Recency (R): Number of days between the snapshot date and the customer's most recent purchase.
- Frequency (F): Number of unique invoices (purchase transactions) made by a customer.
- Monetary (M): Total revenue generated by each customer.

The RFM metrics were computed by grouping the cleaned dataset using CustomerID, resulting in one record per customer.

The resulting RFM dataset formed the input for subsequent clustering and Customer Lifetime Value prediction.

4.6. Machine Learning Methodology

The machine learning component of the project consisted of two stages: customer segmentation and Customer Lifetime Value prediction.

Customer Segmentation

The RFM variables exhibited considerable differences in scale and distribution. Therefore, logarithmic transformation followed by StandardScaler was applied before clustering.

The Elbow Method was used to determine the optimal number of clusters by evaluating the Within-Cluster Sum of Squares (WCSS) for cluster sizes ranging from 1 to 10.

Based on the elbow point, $K = 4$ was selected, and K-Means clustering was performed. The resulting clusters were interpreted and labelled as:

- VIP Customers
- Loyal Customers
- Regular Customers
- At-Risk Customers

Customer Lifetime Value Prediction

Following customer segmentation, Customer Lifetime Value prediction was performed using three supervised machine learning algorithms:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The predictor variables included:

- Recency
- Frequency
- Customer Segment (Label Encoded)

The target variable was Monetary Value, which was used as a proxy for Customer Lifetime Value.

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

using the `train_test_split()` function with a `random_state` of 42 to ensure reproducibility.

4.7. Model Evaluation

The predictive performance of the regression models was assessed using three widely accepted evaluation metrics:

- **Mean Absolute Error (MAE)** – Measures the average magnitude of prediction errors.
- **Root Mean Squared Error (RMSE)** – Measures prediction accuracy while assigning greater penalty to larger errors.
- **Coefficient of Determination (R^2 Score)** – Indicates the proportion of variation in Customer Lifetime Value explained by the model.

The model with the **highest R^2 Score** and the **lowest MAE and RMSE** was considered the best-performing model for this study.

5. DATA ANALYSIS AND RESULTS

This part presents the findings obtained from analysing the Online Retail dataset. The analysis includes descriptive statistics, exploratory data analysis, customer segmentation using K-Means clustering, and Customer Lifetime Value (LTV) prediction using regression models. Visualisations generated during the analysis are used to interpret customer behaviour and evaluate the performance of the machine learning models.

5.1. Descriptive Analysis

Descriptive analysis was performed to understand the characteristics of the customer-level RFM dataset. The RFM analysis converted transaction-level records into customer-level behavioural metrics by calculating **Recency**, **Frequency**, and **Monetary** values for each customer.

	Recency	Frequency	Monetary
count	4338.000000	4338.000000	4338.000000
mean	91.536422	4.272015	2048.688081
std	100.014169	7.697998	8985.230220
min	0.000000	1.000000	3.750000
25%	17.000000	1.000000	306.482500
50%	50.000000	2.000000	668.570000
75%	141.000000	5.000000	1660.597500
max	373.000000	209.000000	280206.020000

Figure 2- Descriptive Statistics (*rfm.describe()*)

Interpretation:

The descriptive statistics reveal substantial variation in customer purchasing behaviour.

1. Recency ranges from customers who purchased very recently to those who have not purchased for several months.
2. Frequency shows that while most customers made only a few purchases, a small proportion purchased repeatedly.
3. Monetary values are highly dispersed, indicating that customer spending varies considerably, with a few customers contributing a significant share of total revenue.

These findings justify the need for customer segmentation to identify groups with similar purchasing behaviour.

5.2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the distribution of the RFM variables and identify the presence of outliers before applying machine learning algorithms.

5.2.1 Histogram Analysis

Histograms were generated for Recency, Frequency, and Monetary variables.

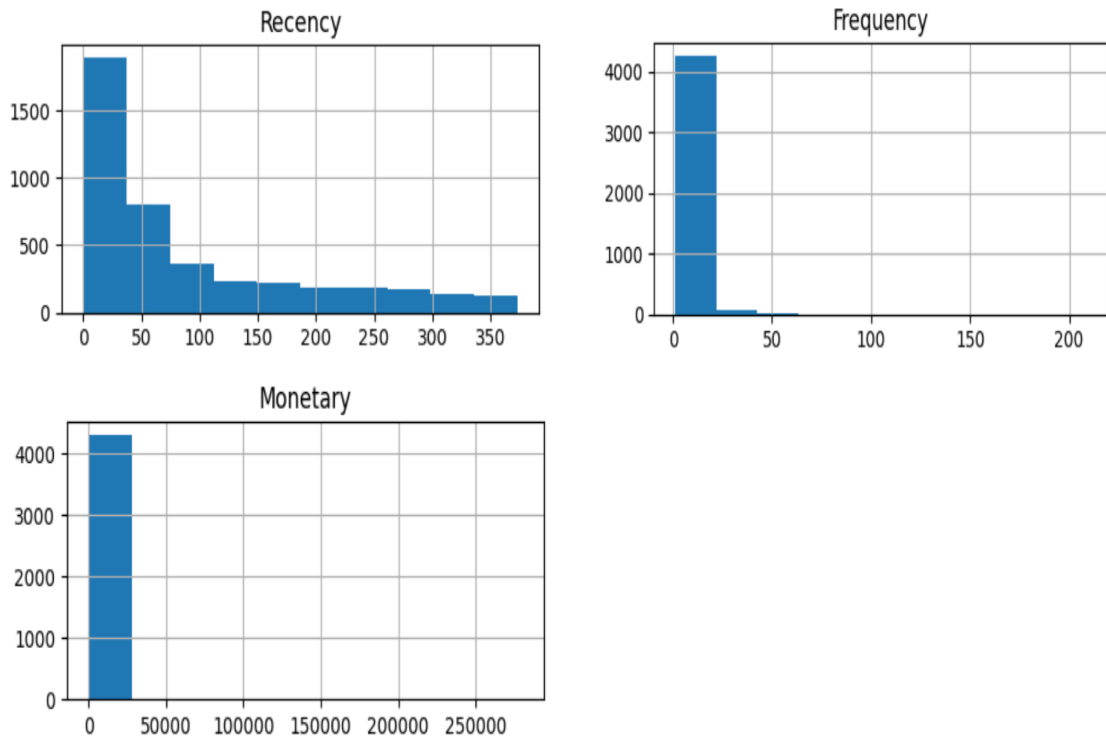


Figure 3- Histogram of RFM Variables

Note:

- **In Recency**, X-axis: Recency (Number of days since the customer's last purchase), Y-axis: Number of Customers (Frequency of observations)
- **In Histogram of Frequency**, X-axis: Frequency (Number of unique purchases/invoices made by a customer), Y-axis: Number of Customers.
- **In Histogram of Monetary**, X-axis: Monetary (Total Revenue Generated by a Customer), Y-axis: Number of Customers.

Interpretation:

The histogram of **Recency** indicates that many customers had purchased relatively recently, while fewer customers exhibited long inactivity periods.

The **Frequency** distribution is highly right skewed, showing that most customers made only a small number of purchases, whereas a limited number of customers purchased frequently.

Similarly, the **Monetary** distribution is strongly right skewed, indicating that most customers contributed relatively low revenue while a few customers generated exceptionally high sales.

These observations demonstrate that the dataset contains highly skewed variables, making preprocessing essential before clustering.

5.2.2 Boxplot Analysis

Boxplots were generated to identify the presence of outliers within the RFM variables.

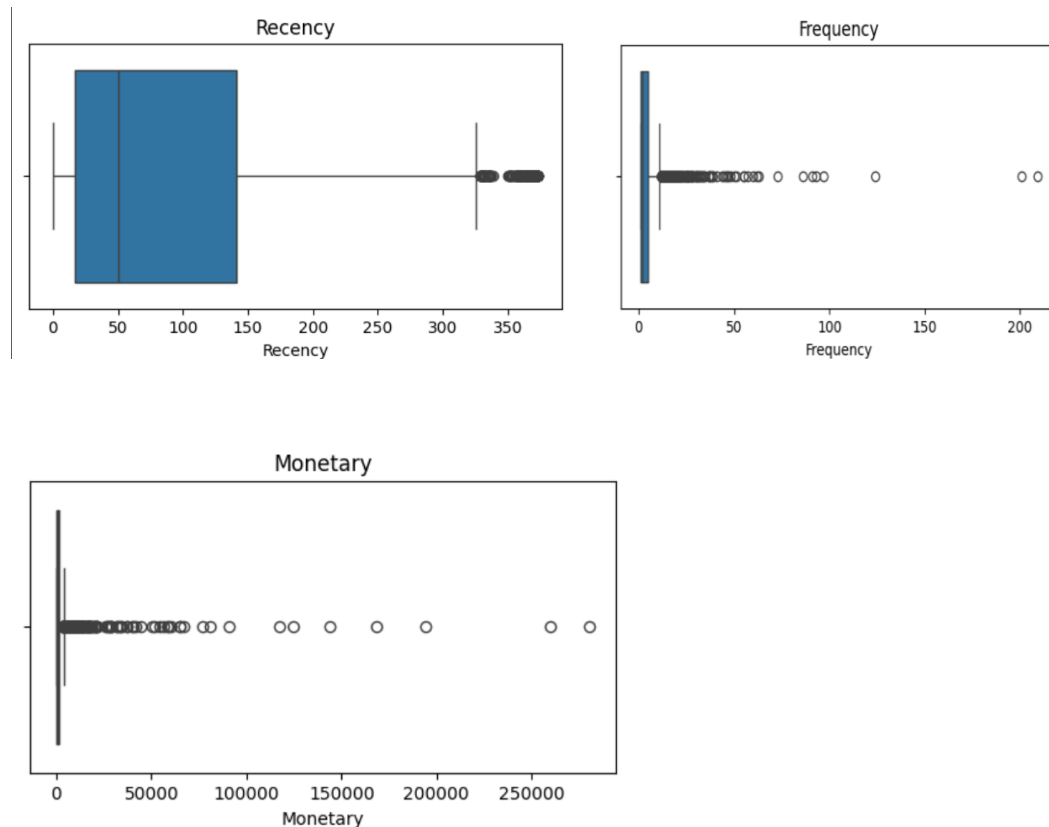


Figure 4- Boxplots of RFM Variables

Interpretation:

The boxplots reveal several extreme observations, particularly in the **Frequency** and **Monetary** variables.

These outliers represent highly valuable customers who make frequent purchases and generate significantly greater revenue than the average customer.

Since K-Means clustering is based on Euclidean distance, such outliers could disproportionately influence the clustering process.

Therefore, logarithmic transformation was applied to reduce skewness while retaining valuable customer information.

5.3. Data Transformation

5.3.1 Log Transformation

A logarithmic transformation ($\log_{10}$) was applied to the Recency, Frequency, and Monetary variables.

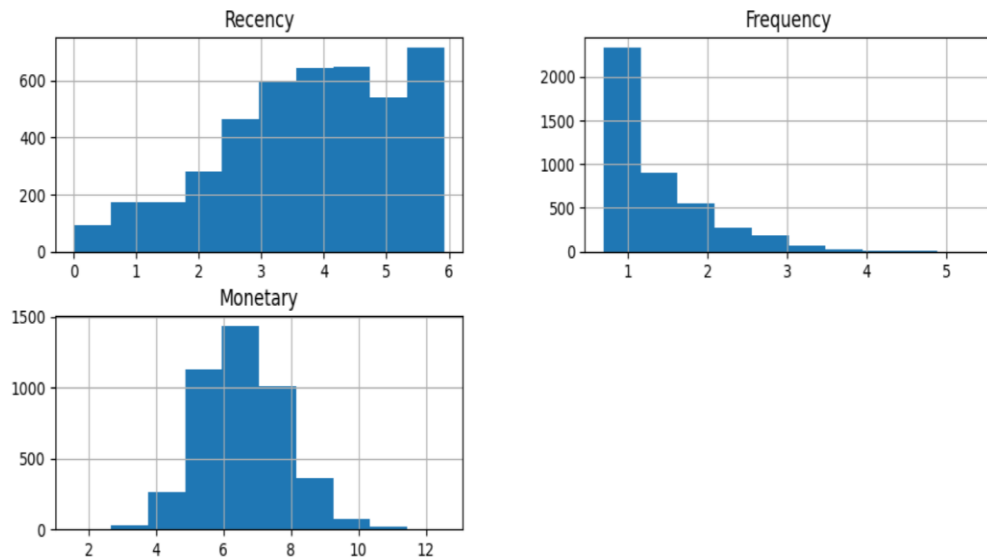


Figure 5- Histogram after Log Transformation

Interpretation:

The transformed distributions became substantially more symmetric compared with the original data.

The logarithmic transformation compressed extreme values while preserving the relative differences between customers. This preprocessing step reduced skewness and produced a more balanced dataset suitable for clustering.

5.3.2 Feature Standardization

Following logarithmic transformation, the variables were standardised using **StandardScaler**.

Standardization transformed each variable to have a mean of approximately zero and a standard deviation of one.

Interpretation:

Since Recency, Frequency, and Monetary are measured using different units and ranges, standardization ensured that each variable contributed equally during distance-based clustering.

Without standardization, variables with larger numerical values would dominate the clustering process.

5.4. Customer Segmentation using K-Means Clustering

5.4.1 Elbow Method

The optimal number of customer clusters was determined using the Elbow Method.

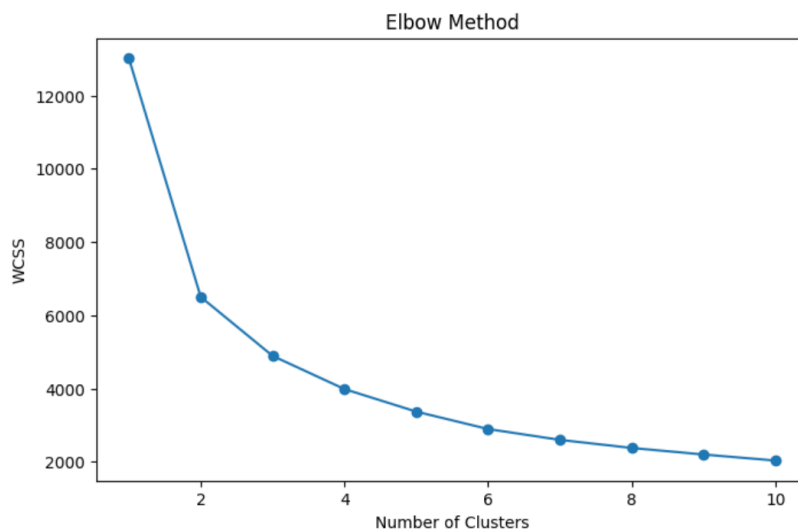


Figure 6- Elbow Method

Interpretation

The Within-Cluster Sum of Squares (WCSS) decreases as the number of clusters increases.

The curve exhibits a noticeable elbow at $K = 4$, indicating that four clusters provide an appropriate balance between clustering quality and model simplicity.

Accordingly, four customer clusters were selected for subsequent analysis.

5.4.2 Cluster Summary

Average Recency, Frequency, and Monetary values were calculated for each cluster to understand customer behaviour.

Table 5.2 – Cluster Summary

	Recency	Frequency	Monetary
Cluster			
0	17.06	2.12	540.35
1	179.43	1.32	347.61
2	10.39	13.99	8290.35
3	67.80	4.21	1829.55

Interpretation:

The cluster summary reveals distinct customer purchasing patterns.

- Customers with **low Recency, high Frequency, and high Monetary values** represent the organisation's most valuable customers.
- Customers exhibiting moderate purchase frequency and spending represent regular purchasing behaviour.
- Customers with long inactivity periods and low spending indicate potential customer attrition.

The clustering process successfully differentiated customers based on purchasing behaviour.

5.4.3 Customer Segment Mapping

The clusters were assigned business-friendly labels based on their purchasing characteristics.

Table 5.3 Customer Segment Description

Segment	Characteristics
VIP Customers	Highest spending and purchase frequency with recent purchases
Loyal Customers	Frequent purchasers with consistently high spending
Regular Customers	Moderate purchasing behaviour
At-Risk Customers	Long inactivity periods with low purchase frequency and spending

5.4.4 Customer Segment Distribution

	Recency	Frequency	Monetary	Cluster	Segment
CustomerID					
12346	325	1	77183.60	3	Loyal Customers
12347	1	7	4310.00	2	VIP Customers
12348	74	4	1797.24	3	Loyal Customers
12349	18	1	1757.55	0	Regular Customers
12350	309	1	334.40	1	At-Risk Customers

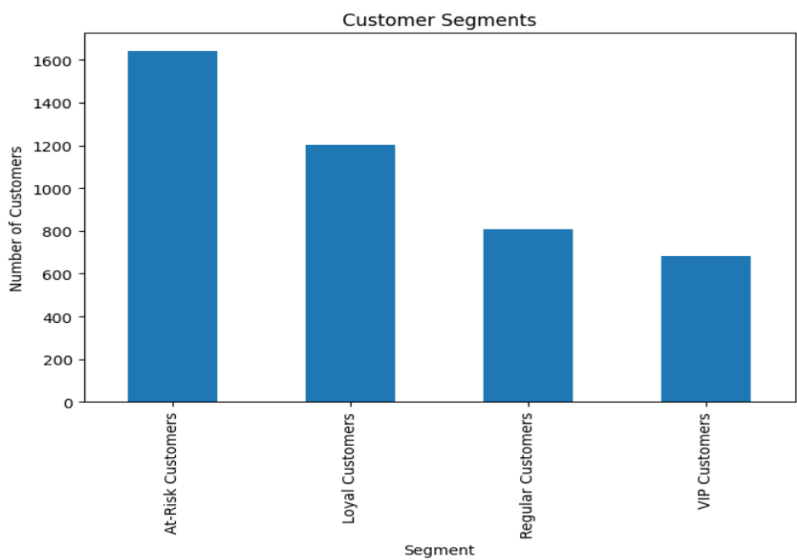


Figure 7- Customer Segment Distribution

Interpretation:

The bar chart illustrates the number of customers belonging to each segment.

The distribution indicates that the customer base consists of multiple behavioural groups requiring different marketing strategies.

VIP Customers contribute substantial revenue and should be prioritised through loyalty programmes and premium services. Loyal Customers should be encouraged to maintain their purchasing behaviour, while At-Risk Customers can be targeted using customer retention campaigns.

5.5. Customer Lifetime Value Prediction

Customer Lifetime Value prediction was performed using three regression algorithms.

- Linear Regression
- Decision Tree Regressor

- Random Forest Regressor

The models were trained using an **80:20 train-test split**, where 80% of the customer records were used for training and the remaining 20% were reserved for testing.

Model performance was evaluated using **MAE**, **RMSE**, and **R² Score**.

5.5.1 Comparative Analysis of Machine Learning Models

Table 5.4 Model Performance Comparison

Model	MAE	RMSE	R ² Score
Linear Regression	1480.51	8209.92	0.3413
Decision Tree Regressor	1985.32	12026.41	-0.4135
Random Forest Regressor	1817.14	10571.48	-0.0922

Interpretation

The comparison indicates that **Linear Regression** achieved the best predictive performance among the three models. It produced the lowest MAE and RMSE values, indicating smaller prediction errors, and achieved the highest R² Score, demonstrating a better fit to the observed customer spending patterns.

The Decision Tree and Random Forest models exhibited negative R² Scores, suggesting that, for the selected features (Recency, Frequency, and Segment), these models did not generalise as effectively as the Linear Regression model.

Consequently, Linear Regression was selected as the most appropriate model for Customer Lifetime Value prediction in this study.

5.6. Summary of Findings

The major findings of the project are summarised below:

- The Online Retail dataset was successfully cleaned and transformed into a customer-level dataset using RFM analysis.
- Exploratory Data Analysis revealed highly skewed customer behaviour and the presence of significant outliers.
- Logarithmic transformation and feature standardisation improved the suitability of the data for clustering.
- The Elbow Method identified four optimal customer segments.

- K-Means clustering successfully classified customers into VIP, Loyal, Regular, and At-Risk groups.
- Customer segmentation provided meaningful insights that can support targeted marketing and customer relationship management.
- Among the evaluated machine learning models, Linear Regression demonstrated the best predictive performance for Customer Lifetime Value estimation.

6. CONCLUSION

This study demonstrated the effective use of customer analytics and machine learning techniques to analyse customer purchasing behaviour and estimate Customer Lifetime Value (LTV) using the UCI Online Retail dataset. The project followed a systematic workflow comprising data cleaning, revenue calculation, RFM feature engineering, exploratory data analysis, customer segmentation, and predictive modelling. By transforming raw transactional data into customer-level behavioural insights, the study established a strong foundation for understanding customer purchase patterns and supporting business-oriented decision-making.

The RFM analysis successfully summarised customer behaviour using Recency, Frequency, and Monetary metrics, while exploratory data analysis highlighted the presence of skewed distributions and outliers that were addressed through logarithmic transformation and feature standardisation. Customer segmentation using the K-Means clustering algorithm identified four meaningful customer groups—VIP, Loyal, Regular, and At-Risk customers—providing valuable insights into customer purchasing behaviour. These segments can assist organisations in designing targeted marketing campaigns, strengthening customer retention strategies, and improving customer relationship management.

For Customer Lifetime Value prediction, three regression models—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—were developed and evaluated using MAE, RMSE, and R^2 Score. Among the evaluated models, **Linear Regression achieved the best predictive performance** for this study by producing the highest R^2 Score and the lowest prediction errors. Overall, the project demonstrates that integrating RFM analysis, customer segmentation, and machine learning provides a practical framework for converting transactional data into actionable business insights. The proposed methodology can support data-driven decision-making in e-commerce and can be extended to other customer-centric industries with similar transactional datasets.

6.1. Limitations of the Study

Although the project achieved its objectives, certain limitations should be acknowledged:

- The analysis was conducted using a single publicly available e-commerce dataset from a UK-based online retailer, which may limit the generalisability of the findings to other industries or geographical regions.
- Customer Lifetime Value was approximated using the Monetary component of RFM analysis, as the dataset did not contain future customer spending information.
- The predictive models were developed using three behavioural features—Recency, Frequency, and Customer Segment. Additional customer characteristics, such as product preferences, purchase intervals, or demographic variables, were not available in the dataset.
- External factors influencing customer purchasing behaviour, such as promotions, seasonality, or economic conditions, were beyond the scope of this study.

6.2. Future Scope

The present study can be extended in several ways to improve customer analytics and predictive performance:

- Incorporate additional customer behavioural features, such as average order value and customer tenure, to improve Customer Lifetime Value prediction.
- Advanced machine learning algorithms, including XGBoost, LightGBM, or CatBoost, may be explored and compared with the current models.
- Interactive dashboards can be developed using Power BI, Tableau, or Streamlit to enable real-time visualisation of customer segments and business insights.
- The proposed methodology can be applied to datasets from other industries, such as banking, healthcare, telecommunications, or retail, to evaluate its effectiveness across different business domains.

7. APPENDICES

Appendix A: References

- <https://trainingthestreet.com/python-packages-101-part-1/>
- https://www.w3schools.com/python/python_modules.asp

Appendix B: GitHub Repository

This link has the datasets and complete code of the project:

- <https://github.com/BavaDharaniBR/Customer-LTV-Prediction>